

Youssuf Helaly

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EXPERIENCE

Software Engineering Intern

Sep. 2026 – Present

Autodesk

Montreal, QC

- Designed and built an **end-to-end AI agent** using LLMs and Model Context Protocol (MCP) to diagnose issues and automate administrative tasks through natural-language, multi-turn workflows.
- Engineered **Java backend services and secure MCP tools**, integrating complex business logic, resource-level authorization, tenant isolation, and deterministic diagnostics for reliable AI-driven administration.
- Analyzed **204** support escalations and partnered with Product, Support, and Engineering to identify and automate high-impact workflows for a channel handling **50** issues daily, reducing manual investigation and support overhead

Software Engineering Consultant (Cloud & ML Infrastructure)

May 2026 – Aug. 2026

ThinkRF

Remote

- Architected an **ML platform processing 50M+ RF observations** from **50+ distributed sensors**, training and deploying **Isolation Forest and LSTM** models for real-time anomaly detection.
- Engineered a **distributed AWS data pipeline ingesting 10M+ records/day** and automated CGI-based enrichment to map global cellular identities across **5 countries** in real time.
- Built an AI engine turning telemetry into automated **root-cause reports**, saving **3+ hours/day** of analyst work

Software Engineering Intern

Sep. 2025 – Dec. 2025

Trend Micro

Ottawa, ON

- Enhanced **distributed Go microservices** deployed on cloud infrastructure by implementing REST APIs and **Prometheus** monitoring, enabling **30%** larger-scale cybersecurity simulations while improving system reliability
- Built a high-throughput pipeline structuring **1,000+** cybersecurity events/attack into indexed datasets for analysis
- Designed and built an **MCP-based environment generator** producing **100+ cybersecurity environments** with dynamic network topologies and embedded CVEs, reducing manual configuration time by **80%**.

Full-Stack Engineering Intern

Jan. 2025 – May 2025

Nokia

Ottawa, ON

- Optimized REST APIs between **React** and **Java Spring** services and fixed service-layer defects, cutting page-load latency by **25%** and improving reliability by **20%**.

Software Engineering Intern

May 2024 – Aug. 2024

Solink

Ottawa, ON

- Engineered **20+ Python REST API endpoints** through a YAML-driven configuration system and built **50+ automated tests**, reducing deployment time by **25%** and manual QA by **30%**.

EDUCATION

University of Ottawa

Expected Dec. 2027

Bachelor of Applied Science in Software Engineering (Co-op) | GPA: 3.95/4.0

Ottawa, ON

PROJECTS

Snug | *Swift, SwiftUI, RealityKit, ARKit, CoreML, TypeScript, three.js*

youssufhelaly.github.io/Snug

- Launched an **iOS app to 25 beta users** that scans rooms on **any iPhone, no LiDAR**, at a **3.4 cm median wall error**, and checks if real Amazon furniture fits via a **±5 cm-band fit engine** with **197 unit tests**.
- Trained a custom **11-class YOLO furniture detector (0.74 mAP50)** deployed **on-device with CoreML**, and cut 3D assets by **96% (682 MB to 28 MB)** with a mesh-simplification pipeline.

ClipIt (Proprietary) | *TypeScript, Rust, TensorFlow, Firebase, React Native*

- Built a cross-platform **React Native video intelligence app** with a **3-stage ML pipeline**, achieving **<500ms inference latency**, **85% detection accuracy**, and real-time processing at **30 FPS**.
- Ran on-device ML via **TensorFlow.js** and **WebGL**, with a **Rust/Actix-Web** backend for FFmpeg clip stitching

TECHNICAL SKILLS

Languages: Python, Go, Java, Rust, TypeScript, JavaScript, SQL

Libraries/Frameworks: Spring Boot, React, React Native, TensorFlow, Scikit-learn, Flask, Next.js, Pandas, NumPy

Tools: AWS, Docker, Kubernetes, PostgreSQL, Redis, Elasticsearch, Firebase, Prometheus, Git, Linux, MCP